

ACP

AARAMBH CHAPTERWISE PROBLEMS

Class-9th

DO NOT CROSS POLICE LINE

DO NOT CROSS POLICE LINE

SOUND

PRASHANT KIRAD



To my Murkh Balaks,

To chalo ye safar bhi khatam hone aa gaya,
aur ye rahi humesha last ACP.

I hope iss journey me tumne padhai ke saath
maje bhi kare honge and bohut kuch seekha
hoga life ke baare me.

Aur ek important baat tum sab mere liye
bohut special ho and I know ki tum apni
life me bohut accha karoge, bohut kuch achieve
karoge, sabko proud feel karoge, I believe in
you all.

I hope sab kuch acche se samajh aaye hoga
Aur haan safar abhi khatam nahi hua
Jo tumse vaada kiya tha ki "ek baar tumhara
haath pakad liya to last tak tumhaare saath
rahunga" vo pura karunga bohut kuch aur
bhi aane wala h aage To wait karna

Till then wish you all, all the best.
Keep shining, keep growing, mehnat krti raho.

- Your Prashant Bhaiya ☺

SOUND

(Options acche se padhna)

MULTIPLE CHOICE QUESTIONS (1 Mark)

1. Sound cannot travel through:

- a) Solids
- b) Liquids
- c) Gases
- d) Vacuum

2. The time taken by a particle to complete one oscillation is called:

- a) Frequency
- b) Wavelength
- c) Time period
- d) Amplitude

3. A sound wave has a frequency of 2 kHz and wavelength of 0.2 m. What is the speed of the wave?

- a) 100 m/s
- b) 200 m/s
- c) 400 m/s
- d) 340 m/s

4. Ultrasound can be used for all except:

- a) Cleaning parts of machines
- b) Breaking stones in kidneys
- c) Detecting gold in rocks
- d) Imaging a fetus in the womb

5. In a longitudinal wave, the particles of the medium move:

- a) Perpendicular to wave direction
- b) In a circular motion
- c) Parallel to wave direction
- d) Not at all

6. The range of audible frequencies for the human ear is:

- a) 20 – 2000 Hz
- b) 2 – 200 Hz
- c) 20 – 20000 Hz
- d) 200 – 20000 Hz

7. Loudness of sound is measured in:

- a) Hertz
- b) Pascal
- c) Decibel
- d) Meter

8. In which type of wave do the particles of the medium vibrate parallel to the direction of wave propagation?

- a) Transverse wave
- b) Longitudinal wave
- c) Electromagnetic wave
- d) Stationary wave

9. A person claps near a cliff and hears the echo after 6 seconds. What is the distance of the cliff from the person?

(Speed of sound = 343 m/s)

- a) 1029 m
- b) 514.5 m
- c) 1715 m
- d) 2058 m

10. A tuning fork produces a sound wave of frequency 512 Hz in air. If the speed of sound is 340 m/s, the wavelength is:

- a) 0.56 m
- b) 0.66 m
- c) 0.72 m
- d) 0.85 m

11. A mechanical piano key struck softly, then again, but much louder this time. In the second case

- a) the sound will be louder, but the pitch will be no different
- b) the sound will be more audible, and the pitch will also be higher
- c) the sound would be louder, but the pitch will be lower
- d) both volume and tone are not affected

12. Which of the following statements about sound waves is true?

- a) Sound can travel through vacuum.
- b) Sound waves are transverse waves.
- c) Sound needs a material medium to travel.
- d) Sound travels faster than light.

13. Which of the following is a correct match?

- a) High amplitude – Low sound
- b) Low frequency – High pitch
- c) High frequency – Shrill sound
- d) High frequency – Loud sound

14. Echo is heard after 0.4 seconds. What is the distance of the reflecting surface?

(Speed of sound = 343 m/s)

- a) 68.6 m
- b) 85.75 m
- c) 102.9 m
- d) 171.5 m

 (yaha Marks katate h)

Assertion And Reasons Questions

A. Both Assertion and Reason are true, and Reason is the correct explanation of Assertion.

B. Both Assertion and Reason are true, but Reason is not the correct explanation of Assertion.

C. Assertion is true but Reason is false.

D. Assertion is false but Reason is true.

15. Assertion: The speed of sound in solids is greater than in gases.

Reason: Solids have higher density than gases.

16. Assertion: We cannot hear the echo produced in a small room.

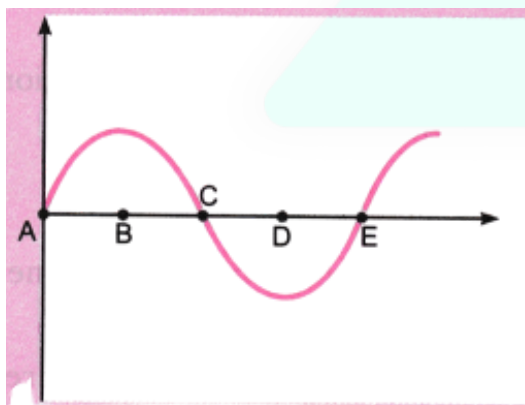
Reason: The reflected sound reaches the ear before 0.1 s and mixes with the original sound.

17. Assertion: In a transverse wave, the particles of the medium vibrate perpendicular to the direction of wave propagation.

Reason: Sound waves in air are transverse in nature.

SHORT ANSWER TYPE QUESTIONS (2 - 3 Marks)

18. Explain how sound is produced by a school bell. Include the role of vibrations.
19. What is meant by the pitch and loudness of sound? On what characteristics of a sound wave do they depend?
20. How does temperature of the medium affect the speed of sound? Briefly explain with reason.
21. What is echo? What is the minimum distance required to hear a distinct echo, assuming speed of sound in air to be 340 m/s ?
22. Explain how sound propagates through air with the help of compressions and rarefactions.
23. What is SONAR? Explain its principle and one use.
24. Why do we hear thunder a few seconds after we see lightning? Explain.
25. What are mechanical waves?
26. What is one complete oscillation?
27. What is the wavelength in the given curve?

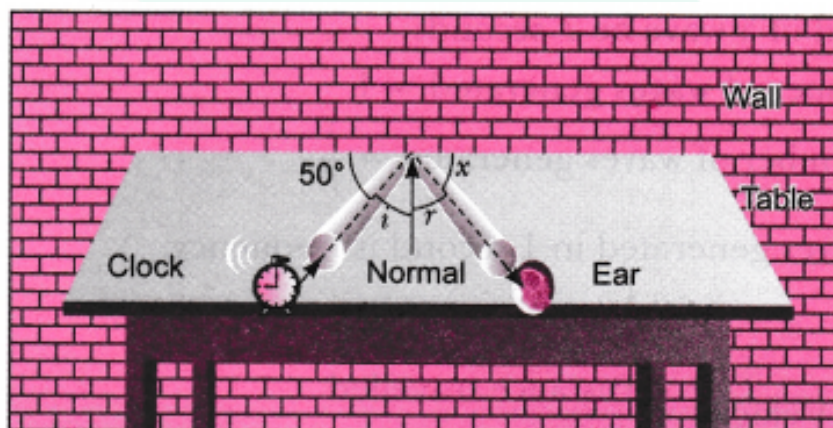


28. Name two animals that communicate using infrasound.

29. Draw a graph for a wave representing wave disturbance and time for a sound changing from low pitch to high pitch, keeping the amplitude of the sound same.

30. Explain the terms crests and troughs of a wave.

31. For hearing the loudest ticking sound heard by the ear, find the angle x in the figure. [NCERT Exemplar]



32. Write the difference between transverse waves and longitudinal waves.

33. Which wave characteristics determine the (a) loudness (b) pitch of sound? Draw two different waveforms and mark these characteristics on it.

34. Plot the following:

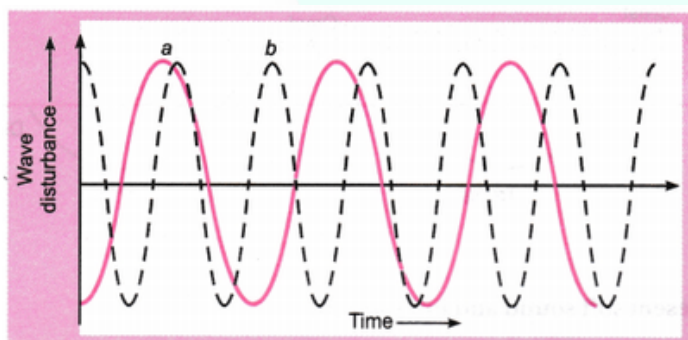
(i) A longitudinal wave in air on a density-distance graph.

(ii) A transverse wave on a displacement-distance graph.

35. Draw diagrams to represent soft sound and loud sound. [CCE 2015]

36. In the graphs given below representing the human voice, which of the two graphs (a) or (b) is likely to be the male voice? Give reason for your answer.

[NCERT Exemplar]



37. A girl is sitting in the middle of a park of dimension $12\text{ m} \times 12\text{ m}$. On the left side of it there is a building adjoining the park and on right side of the park, there is a road adjoining the park. A sound is produced on the road by a cracker. Is it possible for the girl to hear the echo of this sound? Explain your answer. [NCERT Exemplar]

38. Give reasons for the following:

- (a) The reverberation time of a hall used for speeches should be very short.
- (b) A vibrating body produces sound. However no sound is heard when a simple pendulum oscillates in air.
- (c) Sounds of same loudness and pitch but produced by different musical instruments like a violin and flute are distinguishable.

(pahle points socho firr likho)

Long Answer Type Questions (5 Marks)

39. Distinguish between infrasonic, audible, and ultrasonic sounds with frequency ranges. Give one application of ultrasonic sound.

40. Differentiate between transverse and longitudinal waves. Which type of wave is a sound wave and why? Draw neat diagrams of both types.

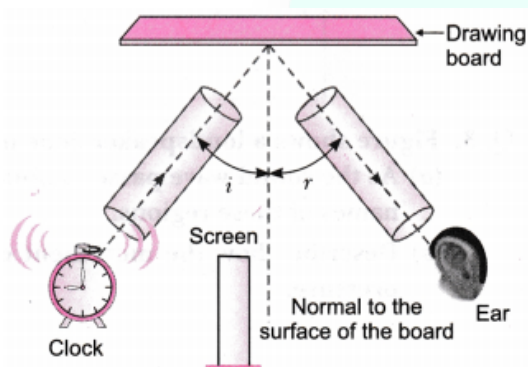
41. What is reverberation? Why is it harmful? Explain two methods used in auditorium design to reduce reverberation.

42. What are the main characteristics of sound? Explain any two of them with suitable examples.

43.(a) What is meant by reflection of sound?

(b) Describe an activity to study the reflection of sound.

(c) State the laws of reflection of sound.



44. Represent graphically any two separate diagrams in each case:

- (i) Two sound waves having the same amplitude but different frequencies.
- (ii) Two sound waves having the same frequency but different amplitudes.
- (iii) Two sound waves having different amplitudes and also different wavelengths. [NCERT Exemplar]

(read paragraph first)

Case Study/Source Based Question

45. Passage:

A concert hall has very hard, flat walls and floor, with little absorbent material. When a musician plays, the sound seems to linger for quite a while even after he stops playing. Some people in the hall complain they cannot understand lyrics properly because the sound overlaps and becomes muddy.

Questions:

- 1. What phenomenon is responsible for the sound lingering?
- 2. Why does this make speech or singing hard to understand?
- 3. Suggest two architectural or material changes to reduce this effect.
- 4. If a sound wave reflects back and forth many times, what happens to its amplitude over time and why?

PK Special [KBC]



46. Why does a guitar string produce a different sound when it is tightened more? Explain in terms of wave properties.

47. A person standing between two tall buildings claps his hands and hears two distinct echoes. Explain how this is possible and under what conditions it happens.

48. The sound of thunder is heard for a longer duration compared to the flash of lightning which is seen instantly. Why does sound last longer, even after it is first heard?

49. Why are stethoscopes designed with flexible tubes filled with air? What property of sound is being utilized here?

50. A person shouts loudly while standing on the surface of a lake. Explain why the sound heard by a person inside the water is different from the one heard in air.

51. Why does sound travel faster on a hot summer day than on a cold winter morning? Give scientific reasoning.

52. Why do we hear more clearly in a closed room than in an open field, even though the source of sound is the same?

53. If astronauts clap their hands on the surface of the moon, why will they not hear any sound even though they are close to each other? Explain scientifically.

54. Figure shows the position of layers of air, at one moment, as a sound wave of constant frequency passes through the air. Compressions are labelled C. Rarefactions are labelled R.



(a) State how figure would change if

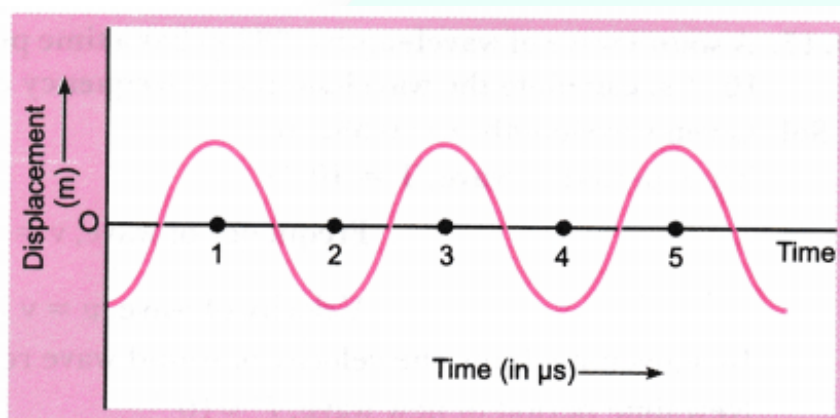
- the sound had a higher frequency,
- the sound were louder.

(b) On figure, draw a line marked with arrows at each end to show the wavelength of the sound.

55. What should be the minimum distance between the listener and the reflector to hear an echo of sound propagating with a speed $v \text{ ms}^{-1}$?

56. The given graph shows the displacement versus time relation for a disturbance travelling with velocity of 1500 ms^{-1} .

Calculate the wavelength of the disturbance. [NCERT Exemplar]



57. A person standing between two vertical cliffs and 640 m away from the nearest cliff shouted. He heard the first echo after 4 seconds and the second echo 3 seconds later. Calculate

- (i) the velocity of sound in air, and
- (ii) the distance between the cliffs

Kitne Questions kar paaye?

SOLUTIONS



SOLUTIONS

MULTIPLE CHOICE TYPE QUESTIONS

1. (d)
2. (c)
3. (c)
4. (c)
5. (c)
6. (c)
7. (c)
8. (b)
9. (a)
10. (b)
11. (a)
12. (c)
13. (c)
14. (a)

ASSERTION AND REASONS TYPE QUESTIONS:

15. (b)
16. (a)
17. (c)

SHORT ANSWER TYPE QUESTIONS

18. When struck, the bell vibrates rapidly.

- These vibrations compress and rarefy the surrounding air particles.
- The compressions and rarefactions travel as a longitudinal wave → we hear sound.

19. Pitch: How “high” or “low” a sound seems; depends on frequency of sound wave.

Loudness: How strong or weak a sound is; depends on amplitude of sound wave (also distance from source).

SOLUTIONS

20. Speed of sound increases with temperature.

- Higher temperature → air molecules move faster → collisions transmit vibrations quickly.
- Example: speed in air at 0°C ≈ 331 m/s; at 20°C ≈ 343 m/s.

21. Echo: Reflection of sound from a hard surface, heard after the original sound.

- To be distinct, time gap ≥ 0.1 s.

$$d = \frac{v \times t}{2} = \frac{340 \times 0.1}{2} = 17 \text{ m}$$

Surface must be at least 17 m away.

22. Vibrating source pushes air → compression (high pressure, density).

- When it moves back → rarefaction (low pressure, density).
- These compressions & rarefactions travel forward, transferring energy as a longitudinal wave without net motion of particles.

23. SONAR: Sound Navigation And Ranging.

- Principle: Ultrasonic waves reflect from underwater objects; time for echo is measured.
- Distance = $v \times t / 2$
- Use: Measuring depth of sea, detecting submarines, locating shipwrecks.

24. Light travels extremely fast (3×10^8 m/s) and reaches us almost instantly. Sound is much slower (≈ 343 m/s), so thunder is heard later, after the lightning is seen.

25. Waves that are characterised by the motion of particles in a medium are called mechanical waves. Mechanical waves require material medium for their propagation.

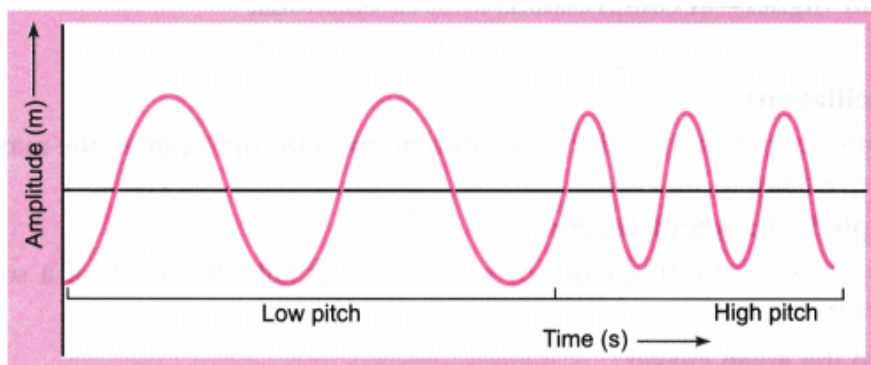
26. The change in density from one maximum value to the minimum value and again to the maximum value makes one complete oscillation.

27. AE

SOLUTIONS

28. Rhinoceroses and whales communicate using infrasound.

29.



30. The elevation or hump in a transverse wave is called crest. It is that part of the transverse wave which is above the line of zero disturbance of the medium. The depression or hollow in a transverse wave is called trough. It is that part of the transverse wave which is below the line of zero disturbance.

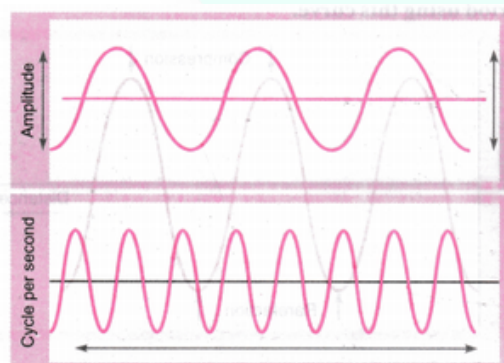
31. According to the laws of reflection, $\angle i = \angle r$

$$\text{So } x = 90^\circ - \angle r = 90^\circ - 50^\circ = 40^\circ$$

32.

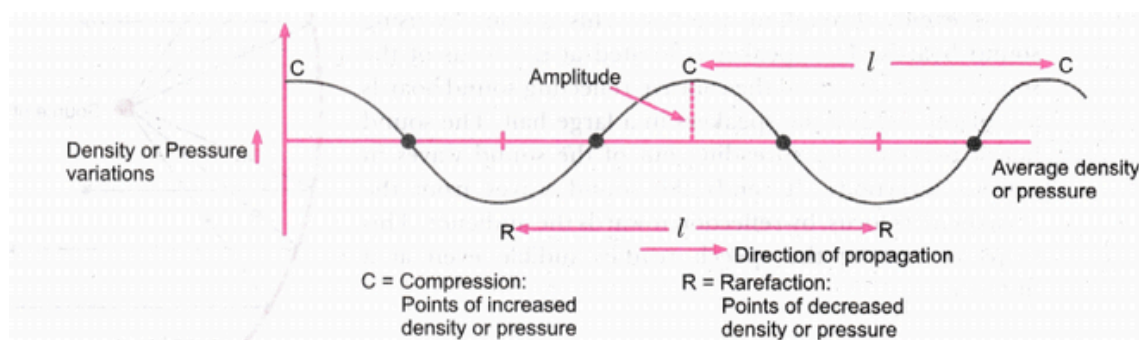
Transverse Waves	Longitudinal Waves
1. The particles of the medium oscillates up and down about their mean position.	1. The particles of the medium move in the parallel to the direction of propagation of the disturbance.
2. They propagate as crests and troughs.	2. They propagate as compressions and rarefactions.
3. The propagation of waves is possible in solid or surface of liquid but not in gases.	3. The propagation of these waves is possible in solids, liquids and gases.
Example: Light wave	Example: Sound wave

33.

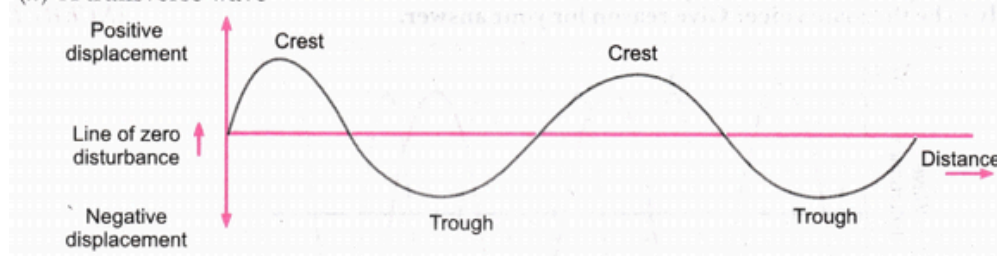


SOLUTIONS

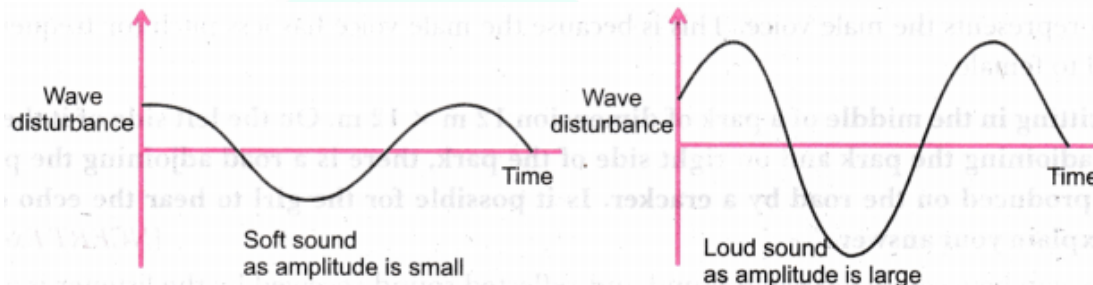
34. (i) A longitudinal wave in air



(ii) A transverse wave



35.



36. Graph (a) represents the male voice. This is because the male voice has less pitch (or frequency) as compared to female.

37. If the time gap between the original sound and reflected sound received by the listener is around 0.1 s, only then the echo can be heard.

The minimum distance travelled by the reflected sound wave for the distinctly listening the echo

$$= \text{Velocity of sound} \times \text{Time interval}$$

$$= 344 \times 0.1 = 34.5 \text{ m}$$

But in this case the distance travelled by the sound reflected from the building and then reaching to the girl will be $(6 + 6) = 12 \text{ m}$, which is much smaller than the required distance. Therefore, no echo can be heard.

SOLUTIONS

38 .(a) If the reverberation time of a hall is long, then the multiple echoes will interfere with the original sound. For this reason, nothing will be heard distinctly. So, the reverberation time of the hall should be very short.

(b) A sound is heard only if the body vibrates with a frequency more than 20 Hz and less than 20,000 Hz. The pendulum oscillates with a frequency less than 20 Hz. Hence, no sound is heard.

(c) This is due to the quality of timbre of sound waves.

39. Infrasonic: Sound waves with frequency below 20 Hz (e.g., elephant communication).

- **Audible:** Sound waves with frequency 20 Hz – 20,000 Hz, heard by human ears.
- **Ultrasonic:** Sound waves with frequency above 20,000 Hz, not audible to humans.
- **Application:** Ultrasonic waves are used in medical imaging (ultrasonography) and in SONAR to detect underwater objects.

40. Transverse waves:

- Particles of the medium vibrate perpendicular to the direction of wave propagation.
- Have crests and troughs.
- Example: Light waves, water waves.
- **Longitudinal waves:**
 - Particles of the medium vibrate parallel to the direction of wave propagation.
 - Have compressions and rarefactions.
 - Example: Sound waves in air.

Sound wave: It is longitudinal because air particles vibrate back and forth along the direction of wave travel.

41.Reverberation: Persistence of sound due to repeated reflections from walls, ceiling, and floor even after the original sound has stopped.

- Harmful effect: Makes speech and music unclear.
- Methods to reduce reverberation:

Use sound-absorbing materials like carpets, curtains, and upholstered seats.

Cover walls and ceilings with porous materials like fiberboards or acoustic panels.

SOLUTIONS

42. The main characteristics of sound are:

1. Loudness

2. Pitch

3. Quality (or Timbre)

1. Loudness: Loudness is the characteristic of sound by which we can distinguish between a loud and a soft sound of the same frequency.

It depends on the amplitude of vibration – the greater the amplitude, the louder the sound produced.

Example: When a drum is beaten hard, it vibrates with a greater amplitude and produces a louder sound than when it is struck gently.

2. Pitch: Pitch is the characteristic of sound that helps us differentiate between a shrill (high-pitched) and a deep (low-pitched) sound.

- It depends on the frequency of vibration – the higher the frequency, the higher the pitch.
- **Example:** The voice of a woman or a child has a higher pitch than that of a man because their vocal cords vibrate faster.

Conclusion:

Thus, loudness depends on the amplitude of vibration, while pitch depends on the frequency of vibration. These characteristics help us identify and distinguish different types of sounds.

43. (a) Reflection is a phenomenon of reversion of a wave going from one medium to the same medium after striking the second medium. Sound wave also experience reflection like other waves.

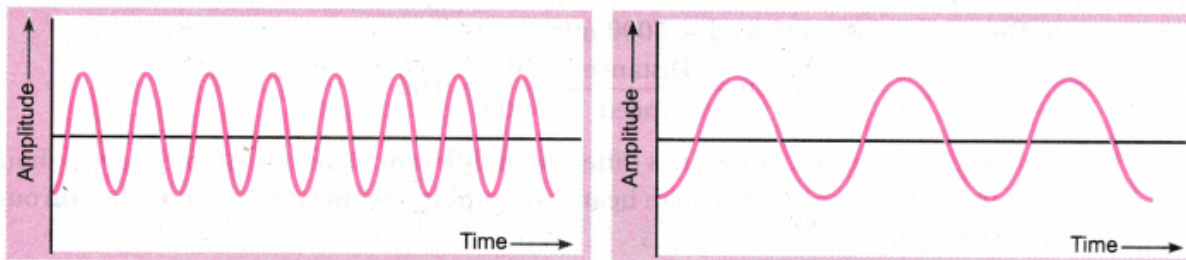
(b) To observe the reflection of sound, take a drawing board and fix it on the floor. Put two metallic or cardboard tubes as shown in figure. These tubes are making some angle with each other. Put a clock near the end of one tube and a screen between the two tubes so that sound of clock may not be heard directly. The sound (like tick-tick) waves pass through the tube are reflected by the drawing board. The reflected sound waves enter the second tube and are heard by the ear placed in front of the second tube.

(c) Laws of reflection:

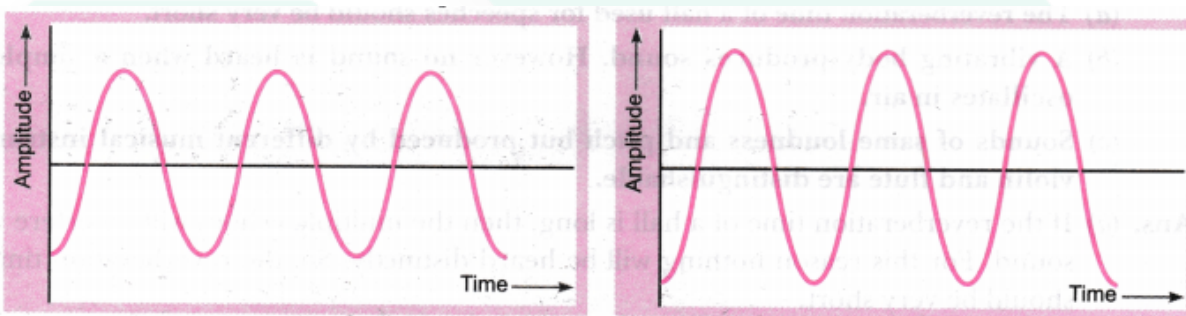
- The angle of incidence is equal to the angle of reflection.
- The incident ray, the normal to the reflecting surface at the point of incidence and the reflected ray, all lie in the same plane.

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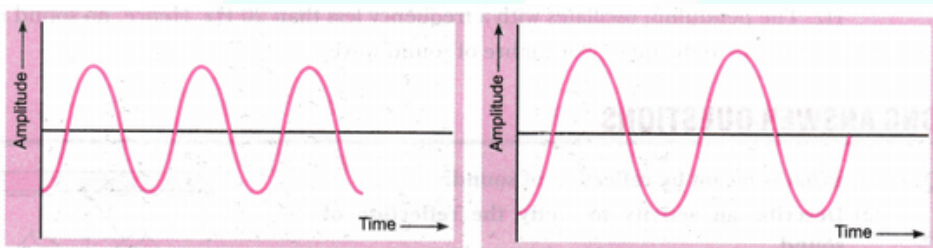
44.(i) Same amplitude but different frequency



(ii) Same frequency but different amplitude



(iii) Different amplitudes and different wavelengths



45. 1. The phenomenon responsible is reverberation – repeated reflections of sound from walls, floor, ceiling etc., causing sound to persist even after source stops.

2. Because many reflected sound waves overlap with newer sounds; this causes “smearing” of sound, reducing clarity. The direct sound and reflected sound mix, making distinct words difficult to hear.

3. Two changes:

- Use absorbent materials (carpets, curtains, upholstered seats) on walls, floor, ceiling to absorb reflections.
- Design curved or non-parallel surfaces so reflections are diffused rather than focusing or repeatedly bouncing directly back.

4. The amplitude gradually decreases over time due to energy loss (absorption by surfaces, air damping). So the reverberated sound gets weaker until inaudible.

SOLUTIONS

PK Special [KBC]

46. Tightening a string increases its tension, which increases the frequency of vibration. Higher frequency produces higher pitch, so the sound becomes shriller.
47. The clap sound reflects separately from each building. If the distances of the two buildings are different enough to cause a time gap of at least 0.1 second, two distinct echoes can be heard.
48. Light travels almost instantly to our eyes, but sound travels slowly. Also, thunder originates from different parts of the cloud at different times and reflects from objects, so the sound reaches us spread out over a longer time, making thunder last longer.
49. Air inside the tube carries sound from the chest piece to the doctor's ears. The property of sound being utilized is transmission of vibrations through a medium.
50. Sound travels faster and more efficiently in water because particles are more closely packed. Thus, a person inside water will hear the sound faster and with different intensity compared to a person in air.
51. At higher temperature, air molecules move faster, so sound waves are transmitted more quickly. Hence, speed of sound increases with temperature.
52. In a closed room, sound waves reflect from walls and reach the listener, making the sound louder and clearer. In an open field, sound spreads in all directions and dissipates quickly, making it weaker.
53. The moon has no atmosphere (no air or medium). Since sound needs a material medium to propagate, it cannot travel in the vacuum of the moon. Hence, astronauts cannot hear the clap.
54. More compressions and rarefactions will be produced in a given time interval (i.e., C and R will become closer together).
At compressions, layers become closer together. At rarefactions, layers become

SOLUTIONS

- farther apart (i.e., C will narrower and rarefactions wider).

(b) Distance between two compressions or two rarefactions are called wavelength (λ)



55.

$$(a) \quad \text{Time} = \frac{\text{Distance}}{\text{Speed}}$$

$$\text{or} \quad t = \frac{2d}{v}$$

$$\text{or} \quad d = \frac{v \times t}{2} = \frac{v \times 1}{2 \times 10} \quad (\because \text{Time} = 0.1 \text{ s}) = \frac{v}{20}$$

56.

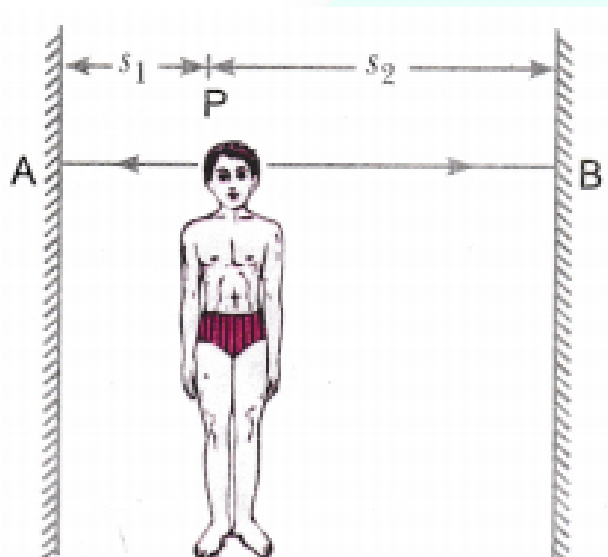
From the graph,

$$\text{Time period, } T = 2 \times 10^{-6}$$

$$\text{Frequency, } \nu = \frac{1}{T} = \frac{1}{2 \times 10^{-6}} = 5 \times 10^5 \text{ Hz.}$$

$$\text{Wavelength, } \lambda = \frac{v}{\nu} = \frac{5 \times 10^5}{1500} = 333.3 \text{ m.}$$

57.



SOLUTIONS

(i) Let P be the person standing between the cliffs A and B. Let s_1 be distance of nearest cliff A from P and s_2 the distance of second cliff B from P. The first echo is heard when sound reaches the person after being reflected from cliff A.

Given, $s_1 = AP = 640$ m

Time interval of first echo, $t_1 = 4$ seconds

From relation, $2s_1 = vt_1$, we have

Speed of sound, $v = 320$ m/s

$\therefore$ Speed of sound in air $v = 320$ m/s

(ii) The second echo is heard when sound reaches the person after being reflected from the cliff B. Time interval of second echo, $t_2 = 4 + 3 = 7$ seconds

$\therefore$ From relation, $2s_2 = vt_2$,

We have, $s_2 = 1120$ m

$\therefore$ Distance between cliffs A and B,

$s = s_1 + s_2 = 640 + 1120 = 1760$ m